E40S6-1000-3-T-24 Datasheet



DiGi Electronics Part Number	E40S6-1000-3-T-24-DG
Manufacturer	Autonics
Manufacturer Product Number	E40S6-1000-3-T-24
Description	ENCODER 40MM 1000PPR 12-24VDC
Detailed Description	Rotary Encoder Incremental 1000 Totem Pole User Selectable

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Manufacturer Product Number:	Manufacturer:
E40S6-1000-3-T-24	Autonics
Series:	Product Status:
E40	Active
DiGi-Electronics Programmable:	Encoder Type:
Not Verified	Incremental
Output Type:	Pulses per Revolution:
Totem Pole	1000
Voltage - Supply:	Actuator Type:
12V ~ 24V	6mm Dia Flatted End
Detent:	Built in Switch:
No	No
Mounting Type:	Orientation:
Chassis Mount	User Selectable
Termination Style:	Rotational Life (Cycles Min):
Cable	-

Environmental & Export classification

RoHS Status:	Moisture Sensitivity Level (MSL):
RoHS Compliant	1 (Unlimited)
REACH Status:	HTSUS:
Vendor Undefined	8536.50.7000

E40 Series

Shaft Type/Hollow Shaft Type/Blind Hollow Shaft Type Ø40mm Incremental Rotary Encoder

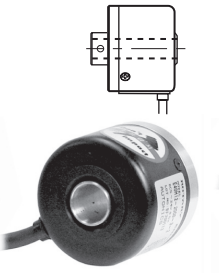
■ Features

- Easy installation at narrow space
- Low moment of inertia
- Power supply: 5VDC, 12-24VDC ±5%
- Various output types

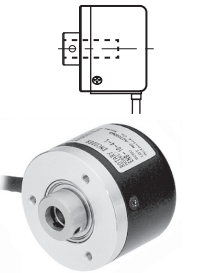
Please read "Safety Considerations" in the instruction manual before using.



E40S Series



E40H Series



E40HB Series

■ Ordering Information

E40 H	8	—	5000	—	3	—	N	—	24	—	

Series	Shaft diameter		Pulses/revolution	Output phase	Control output	Power supply	Cable
Ø40mm S: shaft type	External	6: Ø6mm 8: Ø8mm	Refer to resolution	2: A, B 3: A, B, Z 4: A, $\overline{A}$, B, $\overline{B}$ 6: A, $\overline{A}$, B, $\overline{B}$, Z, $\overline{Z}$	T: Totem pole output N: NPN open collector output V: Voltage output L: Line driver output	5 : 5VDC ±5% 24: 12-24VDC ±5%	No mark : Radial cable type C: Radial cable connector type
Ø40mm H: hollow shaft type, HB: blind hollow shaft type	Inner	6: Ø6mm 8: Ø8mm 10: Ø10mm 12: Ø12mm					

■ Specifications

Item			Shaft Type/Hollow Shaft Type/Blind Hollow Shaft Type Ø40mm Incremental Rotary Encoder
Resolution (PPR)※1			*1, *2 ,*5, 10, *12, 15, 20, 23, 25, 30, 35, 40, 45, 50, 60, 75, 100, 120, 150, 192, 200, 240, 250, 256, 300, 360, 400, 500, 512, 600, 800, 1000, 1024, 1200, 1500, 1800, 2000, 2048, 2500, 3000, 3600, 5000
Electrical specification	Output phase		A, B, Z phase (line driver A, $\overline{A}$, B, $\overline{B}$, Z, $\overline{Z}$ phase)
	Phase difference of output		Phase difference between A and B: $\frac{T}{4} \pm \frac{T}{8}$ (T=1 cycle of A phase)
	Control output	Totem pole output	• [Low] - Load current: max. 30mA, residual voltage: max. 0.4VDC⎓ • [High] - Load current: max. 10mA, output voltage (power voltage 5VDC⎓): min. (power voltage-2.0)VDC⎓, Output voltage (power voltage 12-24VDC⎓): min. (power voltage-3.0)VDC⎓
		NPN open collector output	Load current: max. 30mA, residual voltage: max. 0.4VDC⎓
		Voltage output	Load current: max. 10mA, residual voltage: max. 0.4VDC⎓
		Line driver output	• [Low] - Load current: max. 20mA, residual voltage: max. 0.5VDC⎓ • [High] - Load current: max. -20mA, output voltage (power voltage 5VDC⎓): min. 2.5VDC⎓, Output voltage (power voltage 12-24VDC⎓): min. (power voltage-3.0)VDC⎓
	Response time (rise/fall)	Totem pole output	Max. 1μs (cable length: 2m, I sink = 20mA)
		NPN open collector output	
		Voltage output	
		Line driver output	Max. 0.5μs (cable length: 2m, I sink = 20mA)
	Max. response frequency		300kHz
	Power supply		• 5VDC⎓ ±5% (ripple P-P: max. 5%) • 12-24VDC⎓ ±5% (ripple P-P: max. 5%)
	Current consumption		Max. 80mA (disconnection of the load), line driver output: max. 50mA (disconnection of the load)
	Insulation resistance		Over 100MΩ (at 500VDC megger between all terminals and case)
	Dielectric strength		750VAC 50/60Hz for 1 minute (between all terminals and case)
	Connection		Radial cable type, Radial cable connector type
Mechanical specification	Starting torque		• S type: max. 40gf·cm (0.004N·m) • H/HB type: max. 50gf·cm (0.005N·m)
	Moment of inertia		Max. 40g·cm^2 (4×10^{-6} kg·m^2)
	Shaft loading		Radial: max. 2kgf, Thrust: max. 1kgf
	Max. allowable revolution※2		5,000rpm
Vibration			1.5mm amplitude at frequency of 10 to 55Hz (for 1 min) in each X, Y, Z direction for 2 hours
Shock			Approx. max. 50G
Environment	Ambient temperature		-10 to 70℃, storage: -25 to 85℃
	Ambient humidity		35 to 85%RH, storage: 35 to 90%RH
Protection structure			IP50 (IEC standard)
Cable			Ø5mm, 5-wire (line driver output: 8-wire), 2m, Shield cable (AWG24, core diameter: 0.08, number of cores: 40, insulator out diameter: Ø1mm)
Accessory			• S type: Ø6mm coupling, Ø8mm coupling • H/HB type: bracket
Approval			CE (except line driver output)
Unit weight			Approx. 120g

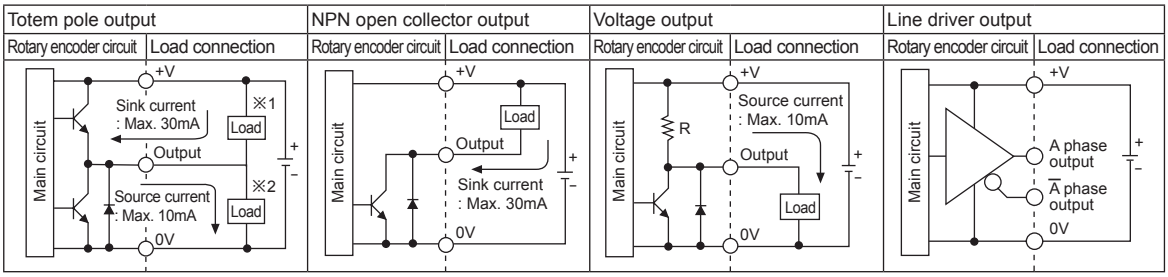
※1: '*' pulse is only for A, B phase (line driver output is for A, $\overline{A}$, B, $\overline{B}$ phase). Not indicated resolutions are customizable.
※2: Make sure that max. response revolution should be lower than or equal to max. allowable revolution when selecting the resolution.

$$[\text{Max. response revolution (rpm)} = \frac{\text{Max. response frequency}}{\text{Resolution}} \times 60 \text{ sec}]$$

※Environment resistance is rated at no freezing or condensation.

Incremental Ø40mm Shaft/Hollow Shaft/Blind Hollow Shaft Type

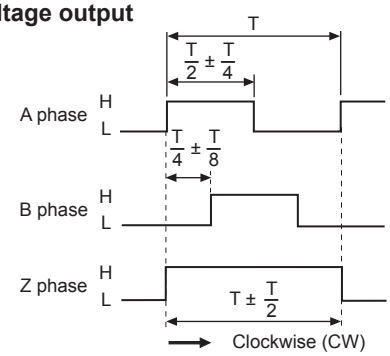
▣ Control Output Diagram



- Totem pole output type can be used for NPN open collector output type (※1) or Voltage output type (※2).
- All output circuits of A, B, Z phase are same. (line driver output is A, Ā, B, B̄, Z, Z̄)

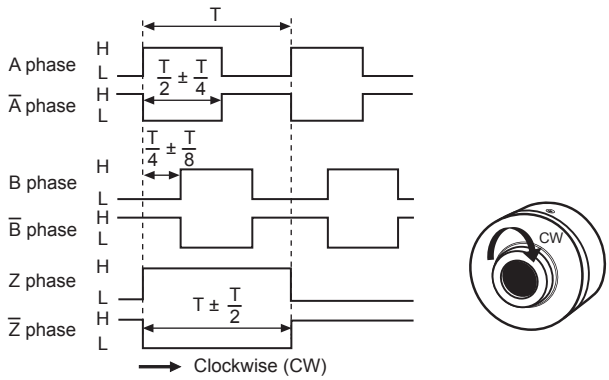
▣ Output Waveform

● Totem pole output / NPN open collector output / Voltage output



※Z reverse phase output is optional.

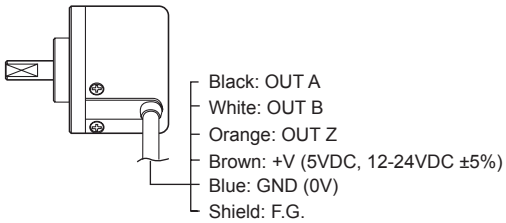
● Line driver output



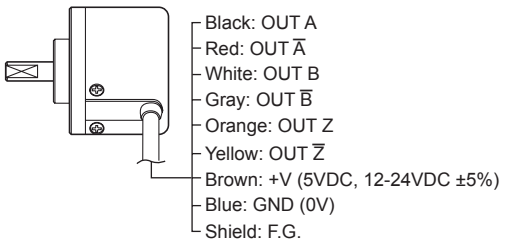
▣ Connections

◎ Radial cable type

● Totem pole output / NPN open collector output / Voltage output



● Line driver output



※Non-using wires must be insulated.
※The shield cable and metal case of encoder must be grounded (F.G.).
※Do not apply tensile strength over 30N to the cable.

◎ Radial cable connector type

● Totem pole output / NPN open collector output / Voltage output



Pin No	Cable color	Function
①	Black	OUT A
②	White	OUT B
③	Orange	OUT Z
④	Brown	+V
⑤	Blue	GND
⑥		F.G.

• Line driver output



Pin No	Cable color	Function
①	Black	OUT A
②	Red	OUT Ā
③	Brown	+V
④	Blue	GND
⑤	White	OUT B
⑥	Gray	OUT B̄
⑦	Orange	OUT Z
⑧	Yellow	OUT Z̄
⑨	Shield	F.G.

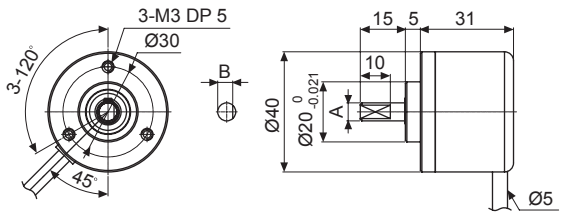
※F.G. (field ground): It should be grounded separately.

E40 Series

▣ Dimensions

(unit: mm)

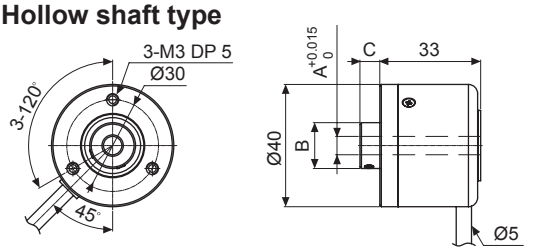
◎ Shaft type



A	Ø6	-0.004 -0.016	Ø8	-0.005 -0.02
B	5		7	

Cable for radial cable type
Ø5mm, 5-wire (line driver output: 8-wire), Length: 2m, Shield cable
Cable for radial cable connector type
Ø5mm, 5-wire (line driver output: 8-wire), Length: 250mm, Shield cable

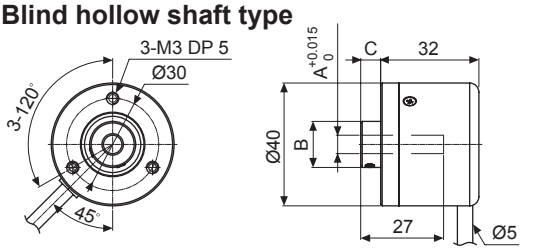
◎ Hollow shaft type



A	Ø6	Ø8	Ø10	Ø12
B	Ø15		Ø17	
C	6.5		6.3	

Cable for radial cable type
Ø5mm, 5-wire (line driver output: 8-wire), Length: 2m, Shield cable
Cable for radial cable connector type
Ø5mm, 5-wire (line driver output: 8-wire), Length: 250mm, Shield cable

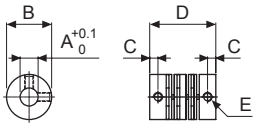
◎ Blind hollow shaft type



A	Ø6	Ø8	Ø10	Ø12
B	Ø15		Ø17	
C	6.5		6.3	

Cable for radial cable type
Ø5mm, 5-wire (line driver output: 8-wire), Length: 2m, Shield cable
Cable for radial cable connector type
Ø5mm, 5-wire (line driver output: 8-wire), Length: 250mm, Shield cable

◎ Coupling (shaft type)

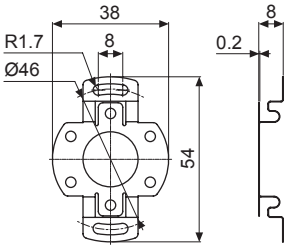


- Parallel misalignment: max. 0.25mm
- Angular misalignment: max. 5°
- End-play: max. 0.5mm

	A	B	C	D	E
E40S6 Ø6mm coupling	Ø6	Ø15	2.8	22	4-M3
E40S8 Ø8mm coupling	Ø8	Ø19	3.4	25	4-M4

※Do not load overweight on the shaft.
※Do not put strong impact when insert a coupling into shaft. Failure to follow this instruction may result in product damage.
※Fix the unit or a coupling by a wrench under 0.15 N·m of torque.
※When you install this unit, if eccentricity and deflection angle are larger, it may shorten the life cycle of this unit.
※For parallel misalignment, angular misalignment, end-play terms, refer to page G-98.
※For flexible coupling (ERB series) information, refer to page G-91.

◎ Bracket (Hollow shaft, blind hollow shaft type)



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